A family of sets over a type that is not a set refutes the truncated axiom of choice

The Univalent Foundations Program

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Editorial note. The counted statement is the author's labelled lemma thm:no-higher-ac at source lines 779-781; its proof environment starts on the line immediately after the environment ends, so the pairing is by adjacency and the retained proof range 782-796 is the whole environment.

Editorial note. Two context spans are retained before the counted statement: the statement of the preceding equivalence (source lines 742-749), because the counted statement refers to the numbered display 3.8.3 that is introduced there and is unreadable without it, and the author's remark at source lines 765-771, which announces the failure of choice for non-sets. Together they also preserve the printed numbering of the section: the retained counters make the context print as Lemma 3.8.2 with display 3.8.3 and Remark 3.8.4, and the counted lemma print as Lemma 3.8.5, exactly as in the book.

Editorial note. Only the labelled lemma at source lines 779-796 is counted; the two context spans are navigation and framing material for the reader and are not counted as propositions.

Editorial note. No context span of a definition is needed: the notions of mere proposition, set and 1-type are standard prerequisites listed above.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-logic/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the truncated axiom of choice fails for a family of sets over a type that is not a set (author label thm:no-higher-ac, original Lemma 3.8.5).

Allowed standard prerequisites (editorial).

- Propositional truncation. The truncation of a type is a mere proposition, and to prove a mere proposition from a truncation it suffices to assume an untruncated witness.. It is what makes the truncation in the statement and the two truncation inductions of the proof (source lines 787 and 794) legitimate.

- Characterisation of paths in Sigma types. A path between pairs is a pair of a path between the first components and a path between the transported second components.. It is used at source line 784 to compute the identity type of the type X.
- Univalence and idtoeqv. Identity of types is equivalent to equivalence of types, so a path between types may be described by an equivalence.. It is used at source line 784 to turn the identity type (A,p) = (B,q) into the type of equivalences A equiv B.
- Mere proposition, set and 1-type. A mere proposition has all its identity types contractible, a set has all its identity types mere propositions, and a 1-type has all its identity types sets.. These three levels are exactly what the proof compares at source lines 785, 787-788 and 795.

Source reference. Original equation 3.8.1: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1206–1206. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original lemma 3.3.3: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1529–1529. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.6.2: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1807–1807. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 3.1.9: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1911–1911. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 3.8.2. *The axiom of choice* (3.8.1) *is equivalent to the statement that for any set X and any $Y : X \to \mathcal{U}$ such that each $Y(x)$ is a set, we have*

$$\Big(\prod_{x:X} \big\|Y(x)\big\|\Big) \to \Big\|\prod_{x:X} Y(x)\Big\|. \tag{3.8.3}$$

Remark 3.8.4. It is easy to show that the right side of (3.8.3) always implies the left. Since both are mere propositions, by original lemma 3.3.3 the axiom of choice is also equivalent to asking for an equivalence

$$\Big(\prod_{x:X} \big\|Y(x)\big\|\Big) \simeq \Big\|\prod_{x:X} Y(x)\Big\|$$

This illustrates a common pitfall: although dependent function types preserve mere propositions (original theorem 3.6.2), they do not commute with truncation: $\left\|\prod_{(x:A)} P(x)\right\|$ is not generally equivalent to $\prod_{(x:A)}\|P(x)\|$. The axiom of choice, if we assume it, says that this is true *for sets*; as we will see below, it fails in general.

Lemma 3.8.5. *There exists a type X and a family $Y : X \to \mathcal{U}$ such that each $Y(x)$ is a set, but such that* (3.8.3) *is false.*

Proof. Define $X :\equiv \sum_{(A:\mathcal{U})}\|\mathbf{2} = A\|$, and let $x_0 :\equiv (\mathbf{2}, |\mathsf{refl}_{\mathbf{2}}|) : X$. Then by the identification of paths in Σ-types, the fact that $\|A = \mathbf{2}\|$ is a mere proposition, and univalence, for any $(A, p), (B, q) : X$ we have $((A, p) =_X (B, q)) \simeq (A \simeq B)$. In particular, $(x_0 =_X x_0) \simeq (\mathbf{2} \simeq \mathbf{2})$, so as in original theorem 3.1.9, X is not a set.

On the other hand, if $(A, p) : X$, then A is a set; this follows by induction on truncation for $p : \|\mathbf{2} = A\|$ and the fact that $\mathbf{2}$ is a set. Since $A \simeq B$ is a set whenever A and B are, it follows that $x_1 =_X x_2$ is a set for any $x_1, x_2 : X$, i.e. X is a 1-type. In particular, if we define $Y : X \to \mathcal{U}$ by $Y(x) :\equiv (x_0 = x)$, then each $Y(x)$ is a set.

Now by definition, for any $(A, p) : X$ we have $\|\mathbf{2} = A\|$, and hence $\|x_0 = (A, p)\|$. Thus, we have $\prod_{(x:X)}\|Y(x)\|$. If (3.8.3) held for this X and Y, then we would also have $\left\|\prod_{(x:X)} Y(x)\right\|$. Since we are trying to derive a contradiction ($\mathbf{0}$), which is a mere proposition, we may assume $\prod_{(x:X)} Y(x)$, i.e. that $\prod_{(x:X)}(x_0 = x)$. But this implies X is a mere proposition, and hence a set, which is a contradiction. □